Parul® University
NAAC A++

BASU SOURJYA SOUMYA

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PROFESSIONAL SUMMARY

My journey in computer science began long before university. From Class 9 onward, I immersed myself in learning how software works—from writing my first programs to exploring advanced computer science concepts through self-driven projects and continuous experimentation.

Over the course of my four-year B.Tech, I expanded this foundation into a broad technical skill set spanning software engineering, embedded systems, operating systems, artificial intelligence, blockchain, backend development, and bare metal programming. My experience ranges from developing web applications and distributed backend systems to building instruction-level emulators, experimenting with operating system development, implementing retrieval-augmented generation (RAG) systems, developing smart contract–based blockchain applications, and writing firmware for embedded devices.

Rather than specializing in a single technology, I have focused on understanding computing across the entire software stack—from hardware-near programming to cloud services and modern AI systems. This breadth enables me to quickly learn unfamiliar technologies, connect ideas across domains, and approach engineering problems from multiple perspectives.

In today's post-AI era, where technology evolves faster than ever, I believe adaptability has become one of the most valuable engineering skills. My diverse technical background allows me to rapidly evaluate new tools, integrate emerging technologies into practical solutions, and contribute wherever the problem demands—whether it involves systems programming, backend engineering, AI, embedded systems, or blockchain.

I am passionate about solving challenging engineering problems, building reliable software, and continuously pushing the boundaries of what I can learn and create.

KEY EXPERTISE

Python | Internet of Things | RTOS | FreeRTOS | Embedded Systems | SQL | Backend Development | Embedded C | SPI | ESP-IDF | Firmware Development | PostgreSQL | Supabase | Firebase | Computer Networks | Socket Programming | Rest API | Flask | Django | FastAPI | RAG | Solidity | Blockchain | Etherium Smart Contracts | PHP | Laravel | NodeJS | Electron | HTML | CSS | Javascript | MongoDB | RAG Systems | RTDB | CISCO Switch Management | Linux System Admin

EDUCATION

Parul University 2023 - 2027
B.Tech. - CSE - PIET | CGPA: **6.87** / **10**

Riverdale Academy 2023
12th | GSHEB | Percentage: **57.54** / **100**

SBV 2021
10th | GSEB | Percentage: **69.83** / **100**

AWARDS AND SCHOLARSHIPS

- Led one of the Top 45 teams among 720 other participating teams in Vadodara Hackathon 6.0

INTERNSHIPS

Czar Metric Pvt. Ltd | IT / Computers - Software 02 Dec, 2025 - 02 Dec, 2025
Intern

Key Skills: Embedded Systems | ESP32

After successfully clearing the interview at Czar Metric Pvt. Ltd. for the Embedded Systems Intern position, I was offered an internship. However, I was unable to accept the opportunity because my university did not issue the required No Objection Certificate (NOC), as internships are not permitted for students in the 5th semester under the university's policy.

PROJECTS

CloudVPN

20 Dec, 2025 - 20 Aug, 2026

Team Size: 4

Key Skills:

Computer Networks | Networking Protocol | Python3 | Protocol Design | Multiplexing | TCP | TCP Tunneling | HTTP | HTTPS | Socket Programming

Project Link: https://github.com/sourjyabose/cloudVPN_client

System architecture that I designed from a bird's eye view:

Browser extension <-> Local client app <-> Server

The design rationale: browser extension can only redirect traffic but cannot process any of it. The main challenge is communication between local client app and server.

So in this project my work focuses on designing and implementing the core networking infrastructure, including protocol design and concurrent data transport systems, till now I built the following:

Developed a custom TCP tunneling and multiplexing system where multiple logical TCP streams are encapsulated, tagged, and routed over one underlying persistent connection to our servers, significantly reducing connection overhead while maintaining isolation between streams.

This system was implemented from scratch using raw sockets, without relying on external networking frameworks or libraries, to achieve fine-grained control over protocol design, data flow, and connection management.

My Key Contributions:

• Designed a custom multiplexed transport protocol with custom binary framing for stream identification and packet routing.

• Implemented full-duplex communication between client, server, and dynamic target endpoints.

• Built a multithreaded architecture with producer–consumer queues for efficient data handling.

• Implemented dynamic socket management with per-stream connection tracking for scalable routing.

• Handled TCP stream fragmentation with custom reassembly logic.

• Integrated with browser extension via PAC scripts for transparent traffic routing.

This project is a collaborative effort, and each team member played a critical role:

Krishna Sharma — thesis development, project report, and research paper.

Roshan — frontend development and authentication server.

Kartik Rohit — cloud infrastructure and asset management.

AVR CPU Emulator

15 Jun, 2026 - 29 Jun, 2026

Team Size: 1

Key Skills:

AVR | Embedded Systems | Python | Assembly Language | Python3 | ISA | Instruction Set | Emulation | Simulator | Computer Architecture

Project Link: https://github.com/sourjyabose/AVRemulator

Built an emulator around ATtiny13 (AVR based). It reads a .bin firmware file and runs it.

• Built an instruction-level AVR (8-bit) microcontroller emulator in Python capable of executing compiled AVR machine code using a fetch–decode–execute architecture.

• Designed a software CPU model implementing a Program Counter (PC), 32 general-purpose registers, Stack Pointer (SPL), Status Register (SREG), RAM, and memory-mapped I/O.

• Developed a mask-and-pattern based instruction decoder, enabling efficient identification and dispatch of AVR opcodes to their corresponding instruction handlers.

• Implemented execution support for multiple AVR instructions including LDI, OUT, IN, RJMP, RCALL, RET, EOR, SBI, SUBI, SBCI, SBIW, BRNE, and NOP, accurately reproducing AVR instruction semantics.

• Simulated AVR stack operations, including function calls and returns, by implementing PUSH/POP mechanisms and return address management.

• Implemented memory-mapped I/O emulation, allowing software peripherals to react dynamically to register writes through callback-based handlers.

• Recreated AVR Status Register (SREG) behavior by computing Carry, Zero, Negative, Overflow, Sign, Half-Carry, and other processor flags according to AVR arithmetic specifications.

• Implemented relative branching and subroutine control flow, including signed offset decoding for branch and jump instructions.

• Engineered a modular emulator architecture separating CPU state, RAM, I/O subsystem, instruction metadata, and execution logic, improving maintainability and extensibility.

• Executed raw AVR binaries directly by reading machine code from binary files and interpreting instructions in real time.

PCD8544 Framebuffer Display Driver Stack For ESP32 (ESP-IDF)

04 Apr, 2026 - 25 Apr, 2026

Key Skills: Embedded C | Embedded Systems | IoT | Internet of Things | Device Driver | ESP-IDF | SPI | RTOS

Project Link: https://github.com/sourjyabose/PCD8544-FRAMEBUFFER-DISPLAY-DRIVER-STACK-FOR-ESP32-ESP-IDF

I have built a zero-dependency 84x48 monochrome LCD device driver on ESP32 for ESP-IDF (A FreeRTOS based SDK) — built 100% from scratch in pure C, no graphic libraries or HAL.

The images below show the public API exposed by my custom-built graphics driver and rendering subsystem.

The driver is available on GitHub: https://lnkd.in/dBM5TZ8b

Current features:
• Hardware-level LCD reset and initialization sequencing using ESP-IDF GPIO APIs
• Low-level command/data transmission pipeline for the PCD8544 controller
• Manual GPIO-clocked serial communication without hardware SPI
• RTOS-compatible LCD timing and synchronization using FreeRTOS delays
• Page-oriented LCD memory addressing for the PCD8544 controller
• Column-major LCD page scanning and update pipeline
• Direct pixel-to-byte monochrome display encoding and bit-packing
• Full 84×48 monochrome framebuffer rendering engine
• Dynamic graphics context and framebuffer attachment abstraction
• Display state management and framebuffer presentation pipeline
• Pixel plotting with boundary-clipped rendering operations
• Rectangle rendering primitives with pixel-level control
• Bresenham-based integer line rasterization
• Midpoint circle rasterization with 8-way symmetry optimization
• Wireframe triangle rendering using line composition
• Bitmap-based 8×8 font rendering engine
• Framebuffer-based text rendering and cursor management system
• Automatic text wrapping and multi-line rendering logic
• Smooth marquee/scrolling text using partial glyph rendering
• Linked-list based coordinate buffering for graphics rasterization, doing transformation on the stored coordinates later.
• Runtime graphics memory management using dynamic allocation
• Lightweight embedded graphics framework written entirely in C without external graphics libraries

Remote Terminal

11 Aug, 2021 - 14 Aug, 2021

Key Skills: NodeJS | bash | Linux | API

I created a NodeJS app which exposed http endpoints. After authentication a user on another computer can send bash commands to it and the app would run it on the server and get the output back in response of the http. If the process took time it will send a url instead and the client side would poll that URL from time to time when completed the output would be in that URL. Once the URL was once fetched the server would then invalidate that URL. I was in Class 11th when i built this project. The dates mentioned are approximate.

KrishiSakhi

Mentor: Jay Sudani | **Team Size:** 6

Key Skills:

Backend Development | Python | RAG | Database | Flask | Streamlit | Flutter | Backend | Memory Management | Python3 | LLM | Large Language Model | Image Recognition | AI Orchestration | AI | AIML | Machine Learning | Linux System Admin | Linux Administration | Web Scrapping | web scraping | API | API Development

Project Link: https://github.com/sourjyabose/crop_advisory_app_server

I built a multi-layer agricultural intelligence and community communication platform where a Flask backend orchestrates conversational AI, ML pipelines, geospatial enrichment, and a persistent hybrid memory system using disk-backed session state and an in-memory LRU cache with timestamped access tracking for continuity.

A dual memory system stores agronomic inputs as state vectors while chat history is transformed into a TF-IDF corpus queried via cosine similarity, injecting only high-confidence matches into the LLM for context.

The decision engine has two paths: crop prediction using a trained ML model on soil, rainfall, and humidity enriched via IP-based city/region geolocation with weather and rainfall data; and fertilizer recommendation using categorical encoding of soil type and crop in a secondary model.

A state-driven routing layer manages chatbot flow and structured questionnaires, while a prompt-conditioned LLM switches between advisory, SMS, and moderation modes with structured UI outputs.

Users are auto-grouped into city-level community channels via IP geolocation, forming persistent localized message ledgers without manual grouping.

A moderation system uses a secondary LLM to flag violations in real time and stores full metadata (user, city, message index).

The admin control plane enables lifecycle governance with warnings, bans, bot restrictions, and unban flows. “comban” controls community access and “botban” restricts AI access, with controlled release restoring flagged messages after review.

A Streamlit admin dashboard provides real-time monitoring of users, communities, chats, and reports with instant ban/unban and message restoration via API mutations.

A news layer scrapes agricultural updates, structures them into headline-link format, and exposes them via API for external intelligence integration.

Vehicle Tracker

Key Skills: Firebase JS JSON Macrodroid JavaScript API Leaflet Real Time Systems

Project Link: https://codepen.io/editor/sourjya-basu/pen/019f4020-d12d-729b-b171-8b8340608843

• Developed a real-time vehicle tracking dashboard using HTML5, CSS3, JavaScript (ES6), and Firebase Realtime Database to monitor and visualize live GPS telemetry. The telemetry data is sent to firebase via a macrodroid script running on drivers mobile.

• Built an interactive geospatial visualization system using Leaflet and OpenStreetMap, supporting route plotting, directional arrows, waypoint markers, and historical path reconstruction.

• Designed an event-driven route selection engine that enables users to interactively select, deselect, and visualize specific GPS data points on the map, allowing customized route reconstruction and analysis.

• Implemented asynchronous data synchronization using the Fetch API, async/await, and timestamp-aware polling to automatically refresh telemetry based on the device's reporting interval.

• Engineered an adaptive exponential backoff and retry mechanism that dynamically increased polling intervals during prolonged device inactivity or communication failures and automatically restored low-latency updates when new telemetry was received, reducing unnecessary network traffic and server load.

• Processed and visualized GPS coordinates, timestamps, battery status, and cellular network metadata (MCC, MNC, LAC, and Cell ID) to provide comprehensive real-time vehicle telemetry monitoring.

• Developed a modular object-oriented frontend architecture for event handling, live tracking, map rendering, and telemetry processing, improving maintainability and code reusability.

• Created a responsive, mobile-friendly interface featuring live monitoring, historical data exploration, customizable route colors, and interactive map controls for efficient fleet tracking.

Unidex - Blockchain based Credential Platform

Team Size: 4

Key Skills:

Blockchain FastAPI Python Python3 Ganache Ethereum Solidity Smart Contracts Android Development Kotlin

Project Link: https://github.com/sourjyabose/PU051_CH214_Special_Ops

Developed a secure blockchain-based backend using FastAPI, MongoDB, and Web3.py for a digital credential platform that enables universities to issue tamper-proof certificates with SHA-256 hashing, smart contract storage on Ethereum, and public verification capabilities. This project was built for participating in hackathon.

• Developed a secure FastAPI backend for a blockchain-based digital certificate verification system (Credential Chain).

• Built RESTful APIs for certificate upload, public verification, and file preview using FastAPI and Pydantic.

• Integrated Ethereum blockchain using Web3.py to store certificate hashes on smart contracts for immutability.

• Implemented SHA-256 hashing for document integrity and duplicate detection during certificate uploads.

• Designed and managed MongoDB schema with PyMongo for storing certificate metadata, student, and university records.

• Developed a JWT-based authentication system for students and universities with role-based access control.

• Created a modular backend architecture with separate layers for routes, services, models, schemas, and utilities.

• Configured and integrated Solidity smart contracts using Hardhat for secure on-chain operations.

• Implemented secure file handling, CORS middleware, and comprehensive error handling for production readiness.

• Built an activity logging service to track certificate uploads and system events.

• Managed environment configuration using dotenv and integrated blockchain connectivity with a local Ganache network.

LetsChat (Now Defunct)

Key Skills: PHP | PhpMyAdmin | Laravel

Project Link: http://suraksha.tekcities.com/

Built a chatting forum using JQuery and JS in frontend and PHP Laravel in backend. It supported images too. I built this project in class 10th or class 11th during lockdown period. I deployed it on a free PHP hosting provider called tekcities.com. This chatting service is now defunct and no longer works due to my account being closed by tekcities.com for prolonged inactivity.

Users joined a community using a community name. If the community didnt existed, it created a new community.
The users then can chat or share images with other fellow users in their community.

PUBLICATIONS / RESEARCH / WHITE PAPERS

A Cost-Efficient TOR-Integrated VPN Architecture with Intelligent Session Management and Load Balancing

Technical Paper | **No. of Authors:** 4

Key Skills: Computer Networking

Problem Statement:- High operational cost on server-side for providing costly single per user VPN sessions and inefficient routing .

Solution:- Intelligent Session Management System , load balancing and sandwich network architecture with TOR integration between entry and exit server.

ACHIEVEMENTS

- Reached the semi-final round of Madhyam, the largest journalism events in Gujarat

ASSESSMENTS / CERTIFICATIONS

CISCO Certified Network Associate

Provider: CISCO Network Academy

Key Skills:

Computer Networking | Computer Networks | Protocols | Networking Protocol | Routing Protocols | DHCP | TCP | UDP

NPTEL Computer Networks

Provider: IIT Kharagpur
Aggregate: 70 / 100
Subjects: Computer Networks (70 / 100)

Key Skills: Computer Networks | TCP

POSITIONS OF RESPONSIBILITY

Placement Coordinator

Selected as a Placement Coordinator through a competitive multi-stage selection process involving group discussions and technical interviews. Coordinated campus recruitment activities by serving as a liaison between recruiters, the placement cell, and students, contributing to the efficient execution of placement drives while demonstrating leadership, communication, and organizational skills.

SEMINARS / TRAININGS / WORKSHOPS

PU DAO - Web3 & Blockchain Hub **Institute Name:** Parul University

Key Skills: Blockchain

• Gained a strong understanding of blockchain consensus mechanisms, including Proof of Work (PoW) and Proof of Stake (PoS), and their trade-offs in security, decentralization, scalability, and energy efficiency.
• Learned Ethereum ecosystem concepts including smart contracts, wallets, transactions, gas fees, and decentralized application workflows.
• Acquired experience with Solidity smart contract development and blockchain development tools such as Hardhat for contract compilation, deployment, and testing.
• Developed familiarity with Web3 integration libraries (Web3.py/Web3.js or ethers.js) for interacting with Ethereum smart contracts.
• Explored blockchain security concepts including cryptographic hashing, digital signatures, transaction verification, and immutable ledger design.
• Participated in collaborative Web3 hackathons, technical workshops, and community-driven blockchain development initiatives

EXTRA CURRICULAR ACTIVITIES

- Journalism
- Anchoring in events

PERSONAL INTERESTS / HOBBIES

- Cycling
- Reading about History and Geopolitics

WEB LINKS

- Github - https://github.com/sourjyabose
- Personal - https://sourjyabose.github.io/

PERSONAL DETAILS

Gender: Male
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Date of Birth: 01 Feb, 2006
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